

Education

- 2021-27 **Ph.D. in Medical Engineering & Medical Physics**
Harvard-MIT Health Sciences & Technology
MIT Electrical Engineering concentration
- Thesis Title: *Asynchronous optical coherence elastography (AsyncOCE) for minimally invasive ophthalmic and intravascular imaging.*
 - Research advisor: Néstor Uribe-Patarroyo
 - Committee members: Brett E. Bouma, Roger Kamm
- 2017-21 **B.S. in Engineering**
Harvard Mudd College
- Research advisors: Christopher Clark, Matthew Spencer

Academic Appointments

- 2021-present **Center for Biomedical OCT Research and Translation (CBORT)**
Graduate student researcher at Massachusetts General Hospital, Harvard Medical School, and Wellman Center for Photomedicine.
Advised by Néstor Uribe-Patarroyo and Brett E. Bouma.
- 2019-21 **Lab for Autonomous and Intelligent Robotics (LAIR)**
Undergraduate researcher at Harvey Mudd College.
Advised by Christopher Clark and Matthew Spencer.

Fellowships

- 2025-26 MIT Health and Life Sciences Collaborative Graduate Fellowship (HEALS)
- 2022-25 National Science Foundation Graduate Research Fellowship (NSF-GRFP)
- 2022-23 MIT Presidential Fellowship
- 2019-21 Jude and Eileen Laspa Fellowship in Autonomous Systems
- 2017-21 Harvey S. Mudd Merit Scholarship
- 2017 National Merit Scholarship

Awards

- 2025 Best Presentation (International Tissue Elasticity Conference)
- 2025 Best Poster Award (SPIE Photonics West, Elastography & Tissue Mechanics)
- 2024 Best Poster Award (Gordon Research Conference, Optics in Medicine)
- 2023 Best Poster Award (Biophotonics Summer School)
- 2022 50 Women in Robotics You Need to Know About (Robohub)
- 2021 Engineering Department Honors (Harvey Mudd College)
- 2021 Johnson Excellence in Engineering Award (Harvey Mudd College)
- 2020 Alford-Gilkeson Award (Harvey Mudd College)

— Publications

- 1) **Ginger Schmidt**, Ryan McAuley, Brett E. Bouma, and Néstor Uribe-Patarroyo, “Asynchronous optical coherence elastography and directional phase gradient analysis,” *Journal of Biomedical Optics* (2025).
- 2) Manisha Singh, Julia Situ, Carly Long, Diego Quevedo-Moreno, **Ginger Schmidt**, Néstor Uribe-Patarroyo and Ellen T. Roche,” *Coronary Artery Flow Modeling in an Ex-vivo Biorobotic Heart*,” 2025 IEEE 8th International Conference on Soft Robotics (2025).
- 3) **Ginger Schmidt**, Brett E. Bouma, and Néstor Uribe-Patarroyo, “Asynchronous, semi-reverberant elastography,” *Optica* 11, 1285-1294 (2024).

— Invited Talks

- 1) SPIE Photonics West in San Francisco, CA: “Towards intraluminal elasticity measurements via an endoscope with asynchronous optical coherence elastography.” (2026)
- 2) MIT Medical Imaging Seminar Series in Cambridge, MA (2026)
- 3) New England Symposium for Biomedical Optics (NESBO) in Cambridge, MA (2025)
- 4) Wellman-University of Tokyo Graduate Student Symposium in Tokyo, Japan (2025)
- 5) Drexel University ECE-101 Guest Lecturer in Philadelphia, PA (2023)

— Contributed Talks

- 1) SPIE Photonics West in San Francisco, CA: “Non-destructive mechanical phenotyping of 3D cartilage tissue models for osteoarthritis with asynchronous optical coherence elastography.” (2026)
- 2) SPIE Photonics West in San Francisco, CA: “Biomechanical assessment of coronary arteries under recapitulated physiological motion with asynchronous shear wave optical coherence elastography.” (2026)
- 3) International Tissue Elasticity Conference in Seattle, WA: “Elasticity measurement of the human cornea in vivo with 3D asynchronous optical coherence elastography.” (2025)
- 4) International Tissue Elasticity Conference in Lyon, France: “Asynchronous semi-reverberant elastography.” (2024)
- 5) SPIE Photonics West in San Francisco, CA: “Overcoming the spatial and temporal coherence limitations of reverberant elastography in raster-scanned OCT systems.” (2023)
- 6) SPIE Photonics West in San Francisco, CA: “Enabling quantitative shear wave elastography in conventional optical coherence tomography in vivo.” (2023)

Industry Positions

- 2020-21 **Johnson & Johnson, Robotics and Digital Surgery**
Mechanical Engineering & Robotics Intern, Laparoscopic surgical robotics.
- 2019 **Raytheon Technologies**
Engineering Intern, Space and Airborne Systems.
- 2018-19 **Auris Health**
Mechanical Engineering & Robotics Intern, Laparoscopic surgical robotics.
Acquired by Johnson & Johnson in 2019.

Supervising & Mentorship

- Isabel Sperandio '25 B.S. Mechanical Engineering, MIT.
Ph.D. Student in Mechanical Engineering at MIT, continuing in our lab.
- Huda Abdelghani '27 B.S. Mechanical Engineering, MIT.
Best Poster at New England Symposium for Biomedical Optics (2025).
- Eric Zou '29 B.S. Electrical Engineering, MIT.

Teaching

- Spring 2021 HMC E206: Robotics State Estimation, TA
- Spring 2021 HMC E102: Advanced Signals & System Engineering 2, Head Grader and Tutor
- Fall 2020 HMC E205: Robotics State Estimation, TA
- Fall 2020 HMC E101: Advanced Signals & System Engineering 2, Head Grader and Tutor
- Fall 2020 HMC E79: Introduction to Systems Engineering, TA
- Spring 2020 HMC E80: Experimental Engineering, Lab Proctor
- Fall 2019 HMC E79: Introduction to Systems Engineering, TA

*HMC = Harvey Mudd College

Service

- 2025-26 Graduate Student Advisory Group for Engineering (GradSAGE): Advisor to the Dean of the MIT School of Engineering, contributing to grad student initiatives.
- 2025 Graduate Admissions Student Interviewer: Interviewed students for admission to the Harvard-MIT Health Sciences & Technology Ph.D. program
- 2021-24 MIT Application Assistance Program: Mentored underrepresented students through graduate school applications.
- 2018-21 Prison Education Project: President and instructor, computer science and physics.

Extracurricular

Season 6 World Championship Winner of Battlebots on the Discovery channel, Captain of combat robotics team Tantrum, appearing on Seasons 3-7.